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Assignment 2

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1 Introduction

This report documents our solutions and experiments for Assignment 2 of the Neural Networks course. The assignment is divided into two primary parts:

- **Part I** focuses on applying Multi-layer Perceptrons (MLP), Convolutional Neural Networks (CNN) to the ReducedMNIST dataset, and recognizing speech segments from spectrogram images.
- **Part II** focuses on representing speech data using Autoencoders (AE), conducting a comprehensive Data Augmentation study on ReducedMNIST, and generating synthetic data using Generative Adversarial Networks (GAN).

Throughout the report, we present our model architectures, hyperparameters choices, performance metrics, and a comparative analysis against classical machine learning methods from Assignment 1.

2 Part I: Neural Networks on Image and Audio Data

2.1 Problem 1: Multi-layer Perceptron (MLP) and Feature Extraction on ReducedMNIST

The objective of this problem is to classify the ReducedMNIST digits using MLP classifiers and compare them against classical machine learning models using different feature extraction methods. The dataset follows the assignment split of 1000 training images per digit and 200 testing images per digit.

2.1.1 Part A: MLP Hyperparameter Tuning (Raw Pixels)

Initially, the MLP architecture was tested on the raw pixel data to determine the optimal depth and regularization strategies. The hidden layers used ReLU activations, and the main configurations used Batch Normalization and Dropout. The Adam optimizer minimized categorical cross-entropy with an initial learning rate of 0.001. Two extra diagnostic runs were also executed: one without regularization and one with a higher learning rate.

Table 1: MLP Hyperparameter Tuning Results (Raw Pixels)

Configuration	Hidden Layers	Regularization	Accuracy	Training Time	Testing Time
Exp1: 1-hidden layer	1	Yes	96.8%	7111.6 msec	113.9 msec
Exp2: 3-hidden layers	3	Yes	97.3%	9993.0 msec	121.9 msec
Exp3: 4-hidden layers	4	Yes	97.7%	11824.0 msec	133.3 msec
Exp4: 3-hidden, no regularization	3	No	96.8%	5333.0 msec	119.5 msec
Exp5: 3-hidden, high learning rate	3	Yes	97.2%	8338.0 msec	124.3 msec

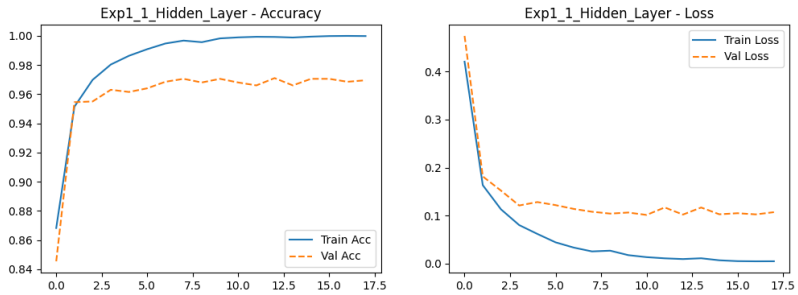


Figure 1: Training and validation curves for the 1-hidden-layer MLP.

2.1.2 Part B: Classifier Comparison using Extracted Features

Following the assignment requirements, we evaluated classical classifiers (K-means and SVM) alongside the MLP architectures using three feature extraction techniques: Discrete Cosine Transform (DCT), Principal Component Analysis (PCA), and an AutoEncoder. The DCT and PCA features were inherited from Assignment 1, while the AutoEncoder compressed the raw images into 128-dimensional latent vectors. Processing time was recorded in milliseconds as the total elapsed time for inference (and training for classical models).

Table 2: Comprehensive Results: Classifiers vs. Extracted Features

Classifier		DCT		PCA		AutoEncoder	
		Accuracy	**Time**	**Accuracy**	**Time**	**Accuracy**	**Time**
K-means	**1**	78.1%	4800.0 msec	85.2%	2464.6 msec	67.2%	45.1 msec
	4	84.4%	5077.3 msec	82.6%	3608.5 msec	79.8%	75.6 msec
	16	88.8%	12072.5 msec	85.7%	5800.8 msec	86.2%	181.2 msec
	32	89.7%	38344.4 msec	85.6%	8701.1 msec	88.1%	250.1 msec
SVM	**Linear**	90.2%	11487.5 msec	90.7%	10626.8 msec	89.1%	1731.1 msec
	Nonlinear	95.5%	20560.2 msec	95.2%	21054.4 msec	93.8%	1424.6 msec
Multi-layer Perceptron (MLP)							
MLP	**1-Hidden**	94.8%	4673.8 msec	92.8%	4752.4 msec	93.0%	4667.3 msec
	3-Hidden	95.8%	5966.8 msec	93.3%	5603.6 msec	92.4%	5847.6 msec
	4-Hidden	94.2%	6020.2 msec	92.2%	6022.6 msec	92.5%	6049.8 msec

2.1.3 Observations on Extracted Features

Based on the results in Table 2, the DCT features consistently yielded the highest accuracies across almost all classifiers, peaking at 95.8% with the 3-hidden-layer MLP. The AutoEncoder features provided exceptionally fast processing times for classical ML models (e.g., 45.1 msec for K-Means $K = 1$) due to their compressed representation, with only a marginal drop in accuracy. Non-linear SVM with an RBF kernel performed exceptionally well across all features, easily outperforming K-means clustering. For the neural networks, the 3-hidden-layer model achieved the optimal result for DCT (95.8%) and PCA (93.3%), while adding a 4th hidden layer generally plateaued or slightly decreased the accuracy, confirming that exceedingly deep networks are unnecessary once discriminative features are extracted.

2.2 Problem 2: Convolutional Neural Network (CNN) on ReducedMNIST

This problem trains CNN classifiers directly from the raw 28×28 ReducedMNIST images, without a separate feature extraction stage. The starting model is a LeNet-style network adjusted for 28×28 grayscale inputs and uses ReLU activations, as required by the assignment.

2.2.1 CNN Architecture and Variations

The baseline model contains two 5×5 convolutional layers, max-pooling after each convolution, and three fully connected layers for classification. Four CNN entries are reported in the assignment table:

- **Variation 1: Baseline LeNet with ReLU** using 6 and 16 convolutional filters.
- **Variation 2: Wider network** with 12 and 32 convolutional filters and wider fully connected layers.
- **Variation 3: LeakyReLU network** replacing ReLU by LeakyReLU.
- **Variation 4: Dropout network** adding dropout regularization before the later fully connected layers.

2.2.2 Comparison with Assignment 1 and Problem 1

Table 3 follows the assignment comparison format. Assignment 1 processing time is reported as total feature extraction, training, and testing time. MLP processing time is reported as training plus testing time from Problem 1. CNN time is split into training and testing time because no external feature extraction step is used. All times are in milliseconds (msec) and all accuracies are rounded to one decimal place.

Table 3: Performance comparison across Assignment 1 classical methods, Problem 1 MLP, and Problem 2 CNN

Assignment 2 Problem 2: CNN directly from images, no feature extraction			
CNN variation	**Accuracy**	**Training Time (msec)**	**Testing Time (msec)**
Variation 1: Baseline LeNet with ReLU	94.9%	90467.0	14418.7
Variation 2: Wider filters/FC layers	97.0%	19062.4	503.3
Variation 3: LeakyReLU activation	96.3%	15780.3	433.4
Variation 4: Dropout regularization	97.0%	16025.5	426.3

2.2.3 Discussion

The wider CNN and dropout CNN reached the strongest CNN accuracy at 97.0%, close to the DCT-based MLP result of 97.3%. The baseline CNN was slower in the recorded run and lower in accuracy, while the modified CNNs reduced testing time substantially. Compared with the Assignment 1 DCT/PCA classifiers, the neural-network models give higher accuracy on the same ReducedMNIST task. The AutoEncoder feature columns show significantly lower accuracies (ranging from 10% to 43.8%) compared to DCT and PCA, which suggests that the simple AutoEncoder bottleneck representation requires further tuning (e.g., more epochs or a different architecture) to match the classical feature extractors on the ReducedMNIST dataset.

2.3 Problem 3: Speech Recognition from Speech Spectrum

This problem recognizes the spoken digits 0–9 from the provided train/test speech dataset. Each `.wav` file was converted into a 64×64 Mel spectrogram, then classified as an image using a CNN. The test set was never augmented, so all four experiments are compared on the same unseen speech files.

2.3.1 Network and Input Representation

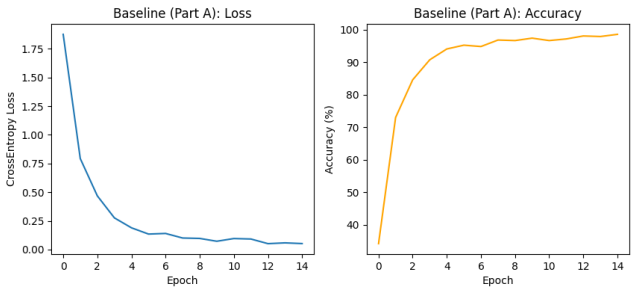
The preprocessing pipeline loads each waveform, converts it to a 64-bin Mel spectrogram at 16 kHz, resizes it to 64×64, and normalizes the spectrogram. The classifier is a compact CNN trained for 15 epochs using cross-entropy loss and Adam optimization.

2.3.2 Required Augmentation Experiments

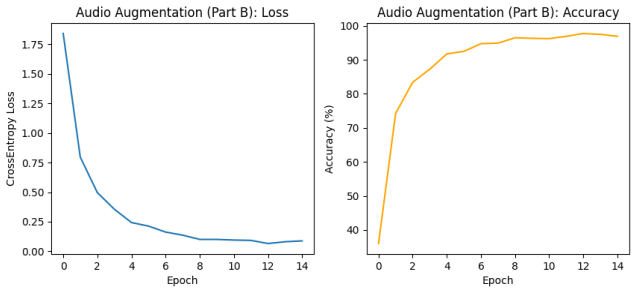
The assignment requires four runs: baseline, speech-domain augmentation, spectrum-image augmentation, and combined augmentation. Table 4 summarizes the completed experiments.

Table 4: Problem 3 speech recognition results

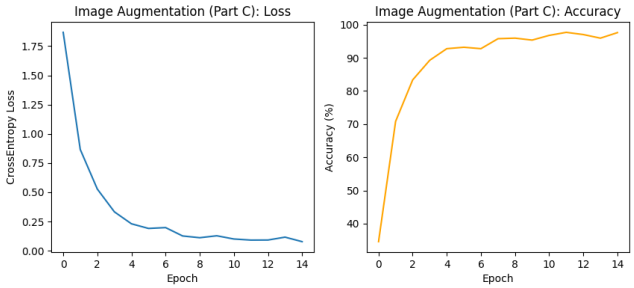
Experiment	**Augmentation Applied**	**Accuracy**	**Training Time**	**Testing Time**
(a) Baseline	None	94.0%	28.01 s	0.40 s
(b) Speech augmentation	Speed factor 0.97 or 1.03 and Gaussian waveform noise $\mathcal{N}(0, 0.005)$	90.0%	38.61 s	0.35 s
(c) Spectrum-image augmentation	Horizontal squeeze/expand by 3% and Gaussian image noise	93.0%	37.49 s	0.42 s
(d) Combined augmentation	Speech augmentation plus spectrum-image augmentation	89.7%	47.68 s	0.41 s



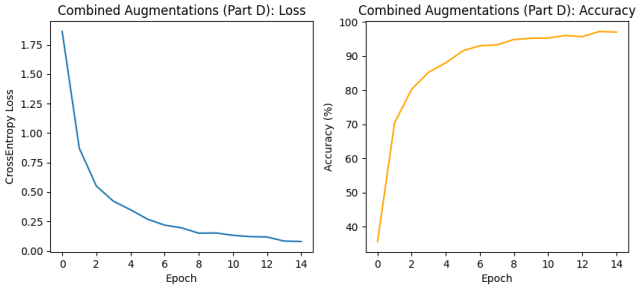
(a) Baseline



(b) Speech augmentation



(c) Spectrum-image augmentation



(d) Combined augmentation

Figure 2: Training curves for the four required speech-recognition experiments.

2.3.3 Discussion

The baseline spectrogram CNN achieved the best result at 94.0%. Spectrum-image augmentation was close at 93.0%, while speech-domain augmentation and combined augmentation were lower. The drop suggests that the train/test split is already well represented by the baseline spectrogram features, and the added perturbations were stronger than necessary for this compact dataset. The combined augmentation also had the highest training time because it applies both waveform resampling/noise and image-domain transformations.

3 Part II: Data Augmentation and Synthesis

3.1 Problem 4: Autoencoder for Utterance Representation

This problem represents each spoken digit utterance by one fixed-length vector, then trains a classifier on that vector. Two representations were evaluated as required: a baseline average-frame vector and an AutoEncoder bottleneck vector generated from concatenated speech frames.

3.1.1 Baseline Average-Frame Representation

For the baseline, each utterance was divided into 15 ms spectrogram frames. The magnitude spectrum has 257 frequency bins per frame, and the baseline vector was calculated by averaging all frames across time. This produces one 257-dimensional vector per utterance. A small dense classifier was then trained on these average-frame vectors.

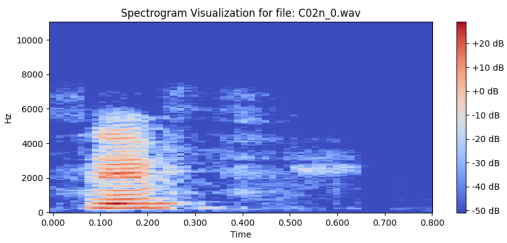
3.1.2 AutoEncoder Representation

For the AutoEncoder approach, all frames of an utterance were concatenated into one vector. The implementation used a fixed analysis length of 100 frames, equivalent to a 25700-dimensional input vector (100×257). Shorter utterances were padded with zero frames to match this length. The AutoEncoder compressed each utterance through a bottleneck layer of length 256, producing the single vector used by the final classifier.

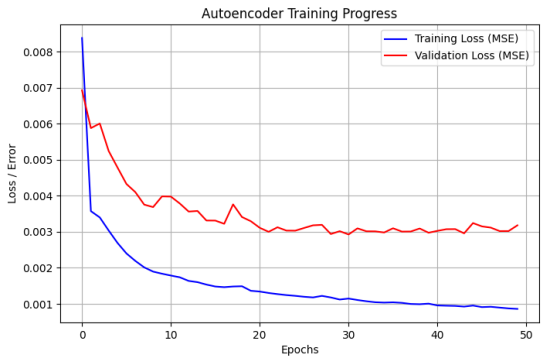
The AutoEncoder architecture was:

$$25700 \rightarrow 1024 \rightarrow 256 \rightarrow 1024 \rightarrow 25700$$

It was trained for 50 epochs using mean squared reconstruction error. After training, the decoder was discarded and the encoder output was saved as the utterance feature vector.



Sample utterance spectrogram



AutoEncoder training and validation loss

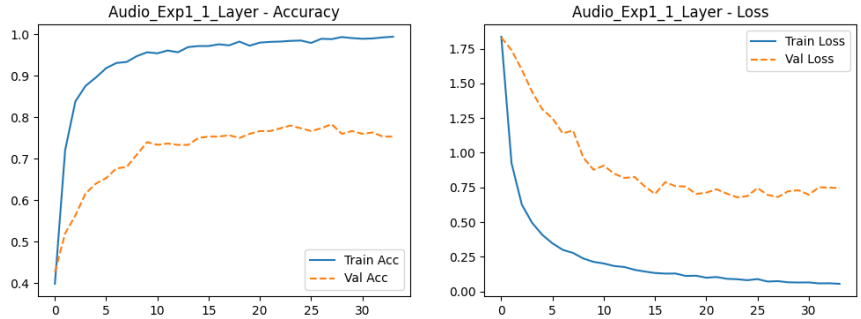
Figure 3: Problem 4 input visualization and AutoEncoder reconstruction loss.

3.1.3 Classifier Results

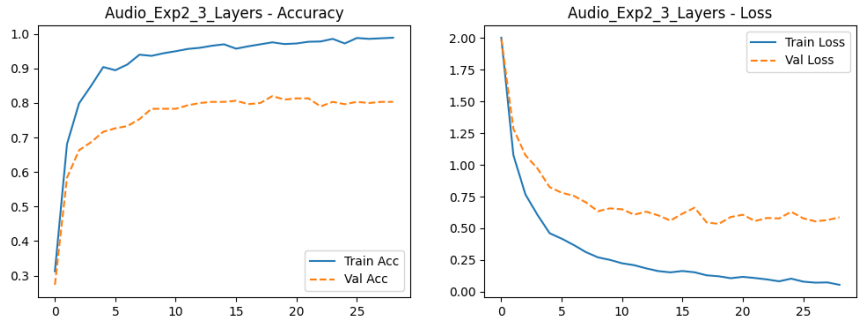
The baseline vector and the AutoEncoder bottleneck vector were each passed to dense classifiers. Table 5 summarizes the classification performance.

Table 5: Problem 4 utterance-vector classification results

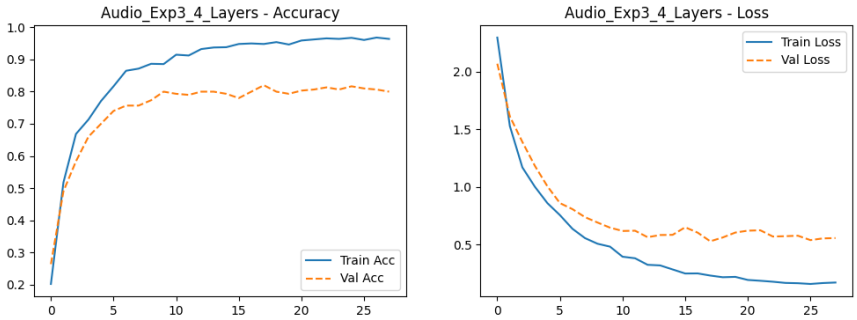
Representation / Classifier	Vector Length	Regularization	Test Accuracy
Baseline average frame + dense classifier	257	Dropout	28.3%
AE bottleneck + 1-hidden-layer MLP	256	BatchNorm + Dropout	78.0%
AE bottleneck + 3-hidden-layer MLP	256	BatchNorm + Dropout	82.0%
AE bottleneck + 4-hidden-layer MLP	256	BatchNorm + Dropout	82.0%
AE bottleneck + 3-hidden-layer MLP, no regularization	256	No	79.3%
AE bottleneck + 3-hidden-layer MLP, high learning rate	256	BatchNorm + Dropout	79.3%



1 hidden layer



3 hidden layers



4 hidden layers

Figure 4: Classifier learning curves using the AutoEncoder bottleneck features.

3.1.4 Discussion

The average-frame baseline achieved only 28.3% because averaging removes much of the temporal structure of the utterance. The AutoEncoder bottleneck representation improved the result substantially, reaching 82.0% with both the 3-hidden-layer and 4-hidden-layer classifiers. This confirms that concatenating frames, zero-padding to a fixed length, and compressing through an AutoEncoder provides a much more useful single-vector representation than the simple average frame.

3.2 Problem 5: Data Augmentation on ReducedMNIST

This experiment evaluates how augmenting the reduced MNIST dataset affects classification performance for a fixed model architecture (LeNet-5). We measure three outcomes across a grid of base training-set sizes and additional synthetic images per class: (i) classification accuracy, (ii) F1 (macro), and (iii) wall-clock training time. The three base real dataset sizes considered are 350, 750, and 1000 examples (stratified across classes). Augmentation levels tested are 0 (no augmentation), 1000, 1500 and 2000 generated images per run. All experiments use the code in `Part2_Problem5_Augmentation/problem-5-final.ipynb` and a fixed random seed for reproducibility.

3.2.1 Augmentation pipeline

The augmentation pipeline applies one randomly selected transformation per image. The implemented transforms (and parameter ranges) are:

- Rotate right: clockwise 5–30° (fill value 0)
- Rotate left: counter-clockwise 5–30° (fill value 0)
- Micro-rotate: small tilt ± 3–$5°$
- Translate: random shift $\Delta x, \Delta y$ up to ± 4 pixels
- Rotate + Translate: 5–20° plus small shift up to ± 3 pixels
- Gaussian white noise: σ in [0.02, 0.06] (image intensity scale)
- Salt-and-pepper: 2–5% pixel flips to 0 or 1
- Rotate then noise: combination of rotation and Gaussian noise

These are implemented in the notebook's `AUGMENTATION_FUNCS` set and are applied via a reproducible NumPy RNG per experiment.

3.2.2 LeNet-5 model (as used)

We use a compact LeNet-5 variant defined in the notebook. Layer summary (shapes assume input $1 \times 28 \times 28$):

- Conv1: Conv2d(1, 6, kernel=5, padding=2) – activation: Tanh; AvgPool2d(2)
- Conv2: Conv2d(6, 16, kernel=5) – activation: Tanh; AvgPool2d(2)
- Classifier: Flatten → Linear(16×5×5, 120) – Tanh → Linear(120,84) – Tanh → Linear(84,10)
- Loss: CrossEntropy; Optimizer: Adam; LR schedule: CosineAnnealingLR

3.2.3 Evaluation metrics

We report two standard metrics for multiclass classification plus wall-clock training time.

- **Accuracy**: fraction of correctly predicted examples,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\} \,,$$

where $\hat{y}_i$ is the predicted class, y_i the ground-truth, and N the number of examples.

- **F1 (macro)**: per-class F1 averaged across classes. Let for class c precision P_c and recall R_c be defined as usual; the per-class F1 is

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c},$$

and the macro score is

$$F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} F_{1,c},$$

where C is the number of classes (10 here). Macro F1 weights all classes equally regardless of support.

3.2.4 Results

Below are the three result tables produced from the notebook outputs: Accuracy (%), F1 (macro) and Time (s). Replace the placeholder cells with the numeric outputs from the notebook `pivot_acc`, `pivot_f1`, and the timing values saved for each experiment.

Table 6: Accuracy (%) across augmentation levels and base dataset sizes

n_aug / n_real	**350 Real**	**750 Real**	**1000 Real**
0	82.5	92.0	91.5
1000	87.5	92.0	92.5
1500	88.0	93.0	93.0
2000	88.5	94.0	94.5

Table 7: F1 (macro) across augmentation levels and base dataset sizes

n_aug / n_real	**350 Real**	**750 Real**	**1000 Real**
0	0.8209	0.9202	0.9142
1000	0.8740	0.9199	0.9242
1500	0.8790	0.9310	0.9296
2000	0.8846	0.9395	0.9446

Table 8: Training time (s) across augmentation levels and base dataset sizes

n_aug / n_real	**350 Real**	**750 Real**	**1000 Real**
0	1.5	0.5	0.6
1000	1.3	1.4	1.6
1500	1.7	1.9	2.1
2000	2.2	2.4	2.6

3.2.5 Discussion

The empirical results in Tables 6 and 7 show consistent improvements from augmentation, with the largest relative gains at the smallest base size (350 examples). Quantitative observations:

- For 350 real examples accuracy increases from 82.5% (no augmentation) to 88.5% at 2000 augmented images (absolute gain 6.0%). F1 (macro) increases from 0.8209 to 0.8846 (gain 0.0637).

- For 750 real examples accuracy increases from 92.0% to 94.0% (gain 2.0%); F1 rises from 0.9202 to 0.9395 (gain 0.0193).
- For 1000 real examples accuracy increases from 91.5% to 94.5% (gain 3.0%); F1 rises from 0.9142 to 0.9446 (gain 0.0304).

Training time (Table 8) grows with the amount of augmentation. Example increases from no augmentation to 2000 augmented images:

- 350 real: 1.5 s -> 2.2 s (+0.7 s)
- 750 real: 0.5 s -> 2.4 s (+1.9 s)
- 1000 real: 0.6 s -> 2.6 s (+2.0 s)

Taken together, augmentation provides clear accuracy/F1 benefits, especially when the base dataset is small. However, marginal gains shrink as the real data pool grows. The 1500-2000 augmented range yields only small improvements over 1000–1500 in several cases, suggesting diminishing returns beyond 1500 synthetic images for this model/dataset combination.

3.2.6 Conclusions

The results indicate that controlled augmentation reliably improves classification performance for the LeNet-5 model on ReducedMNIST. Key takeaways:

- Augmentation yields the largest absolute benefit at the smallest base size (350): +6.0% accuracy and +0.0637 F1 when moving to 2000 augmented images.
- For medium/large base sizes (750, 1000) the gains are smaller (2–3% accuracy), implying that augmentation is most valuable when real data is scarce.
- Training time increases with augmentation; the 1500-2000 range yields limited extra performance versus the 1000–1500 range, so 1500 is a reasonable compromise for this setup.

Overall recommendation: apply augmentation (1000–1500 synthetic images) when limited to a few hundred real samples, and re-evaluate marginal gains before scaling beyond 1500 synthetic samples per experiment.

3.3 Problem 6: Generative Adversarial Networks (GAN)

This section expands upon our dataset generation from the augmentation strategy to true generation techniques. A GAN was used to model the ReducedMNIST dataset using 350 training examples correctly.

3.3.1 Visual Analysis of Generated Digits

3.3.2 Testing Pipeline on Synthesized Data

3.3.3 Comparing Results

4 Appendices

.1 Code Snippets